

Striver's A2Z DSA Sheet

Learn DSA from A to Z

474

TOTAL PROBLEMS

151

EASY

187

MEDIUM

136

HARD

260

WITH LEETCODE LINK

Source: <https://takeuforward.org/dsa/strivers-a2z-sheet-learn-dsa-a-to-z> · compiled 02 September 2026 · problem titles link to the LeetCode problem where one exists; otherwise the row links to the takeUforward article for that problem.

Learn the basics

54 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Things to Know in C++/Java/Python or any language			
1	Input Output	Easy	no LeetCode · takeuforward.org
2	Cpp Basics	Easy	no LeetCode · takeuforward.org
3	If Elseif	Easy	no LeetCode · takeuforward.org
4	Switch Case	Easy	no LeetCode · takeuforward.org
5	What are arrays, strings?	Easy	no LeetCode · takeuforward.org
6	For loops	Easy	no LeetCode · takeuforward.org
7	While loops	Easy	no LeetCode · takeuforward.org
8	Functions (Pass by Reference and Value)	Easy	no LeetCode · takeuforward.org
9	Theory with examples	Easy	no LeetCode · takeuforward.org
Build-up Logical Thinking			
10	Easy and Medium	Easy	no LeetCode · takeuforward.org
11	Hard	Easy	no LeetCode · takeuforward.org
Patterns			
12	Pattern 1	Easy	no LeetCode · takeuforward.org
13	Pattern 2	Easy	no LeetCode · takeuforward.org
14	Pattern 3	Easy	no LeetCode · takeuforward.org
15	Pattern 4	Easy	no LeetCode · takeuforward.org
16	Pattern 5	Easy	no LeetCode · takeuforward.org
17	Pattern 6	Easy	no LeetCode · takeuforward.org
18	Pattern 7	Easy	no LeetCode · takeuforward.org
19	Pattern 8	Easy	no LeetCode · takeuforward.org
20	Pattern 9	Easy	no LeetCode · takeuforward.org
21	Pattern 10	Easy	no LeetCode · takeuforward.org
22	Pattern 11	Easy	no LeetCode · takeuforward.org
23	Pattern 12	Easy	no LeetCode · takeuforward.org
24	Pattern 13	Easy	no LeetCode · takeuforward.org
25	Pattern 14	Easy	no LeetCode · takeuforward.org

#	PROBLEM	LEVEL	LEETCODE LINK
26	Pattern 15	Easy	no LeetCode · takeuforward.org
27	Pattern 16	Easy	no LeetCode · takeuforward.org
28	Pattern 17	Easy	no LeetCode · takeuforward.org
29	Pattern 18	Easy	no LeetCode · takeuforward.org
30	Pattern 19	Easy	no LeetCode · takeuforward.org
31	Pattern 20	Medium	no LeetCode · takeuforward.org
32	Pattern 21	Medium	no LeetCode · takeuforward.org
33	Pattern 22	Medium	no LeetCode · takeuforward.org

Learn STL/Java-Collections or similar thing in your language

34	STL	Easy	no LeetCode · takeuforward.org
35	Java Collections	Easy	no LeetCode · takeuforward.org

Know Basic Maths

36	Count all Digits of a Number	Easy	no LeetCode · takeuforward.org
37	Reverse a number	Easy	leetcode.com/problems/reverse-integer
38	Palindrome Number	Easy	leetcode.com/problems/palindrome-number
39	GCD of Two Numbers	Easy	no LeetCode · takeuforward.org
40	Check if the Number is Armstrong	Easy	no LeetCode
41	Print all Divisors	Easy	no LeetCode · takeuforward.org
42	Check for Prime Number	Easy	no LeetCode · takeuforward.org

Learn Basic Recursion

43	Understand recursion by print something N times	Easy	no LeetCode · takeuforward.org
44	Print name N times using recursion	Easy	no LeetCode · takeuforward.org
45	Print 1 to N using Recursion	Easy	no LeetCode · takeuforward.org
46	Print N to 1 using Recursion	Easy	no LeetCode · takeuforward.org
47	Sum of First N Numbers	Easy	no LeetCode · takeuforward.org
48	Factorial of a given number	Easy	no LeetCode · takeuforward.org
49	Reverse an array	Easy	no LeetCode · takeuforward.org
50	Check if String is Palindrome or Not	Easy	leetcode.com/problems/valid-palindrome
51	Fibonacci Number	Easy	leetcode.com/problems/fibonacci-number

Learn Basic Hashing

52	Basic Hashing	Easy	no LeetCode · takeuforward.org
53	Counting Frequencies of Array Elements	Easy	no LeetCode · takeuforward.org
54	Highest Occurring Element in an Array	Easy	leetcode.com/problems/frequency-of-the-most-frequent-element

Learn Important Sorting Techniques

7 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Sorting-I			
55	Selection Sort	Easy	no LeetCode · takeuforward.org
56	Bubble Sort	Easy	no LeetCode · takeuforward.org

#	PROBLEM	LEVEL	LEETCODE LINK
57	Insertion Sorting	Easy	no LeetCode · takeuforward.org

Sorting-II

58	Merge Sorting	Medium	no LeetCode · takeuforward.org
59	Recursive Bubble Sort	Easy	no LeetCode · takeuforward.org
60	Recursive Insertion Sort	Easy	no LeetCode · takeuforward.org
61	Quick Sorting	Easy	no LeetCode · takeuforward.org

Solve Problems on Arrays [Easy -> Medium -> Hard]

40 problems

#	PROBLEM	LEVEL	LEETCODE LINK
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Easy

62	Largest Element	Easy	no LeetCode · takeuforward.org
63	Second Largest Element	Easy	no LeetCode · takeuforward.org
64	Check if the Array is Sorted II	Easy	leetcode.com/problems/check-if-array-is-sorted-and-rotated
65	Remove duplicates from Sorted array	Easy	leetcode.com/problems/remove-duplicates-from-sorted-array
66	Left Rotate Array by One	Easy	leetcode.com/problems/rotate-array
67	Left Rotate Array by K Places	Easy	leetcode.com/problems/rotate-array
68	Move Zeros to End	Easy	leetcode.com/problems/move-zeroes
69	Linear Search	Easy	no LeetCode · takeuforward.org
70	Union of two sorted arrays	Easy	no LeetCode · takeuforward.org
71	Find missing number	Easy	no LeetCode · www.geeksforgeeks.org
72	Maximum Consecutive Ones	Easy	leetcode.com/problems/max-consecutive-ones
73	Find the number that appears once, and other numbers twice.	Medium	leetcode.com/problems/single-number
74	Longest subarray with given sum K(positives)	Medium	no LeetCode · takeuforward.org
75	Longest subarray with sum K	Medium	no LeetCode · takeuforward.org

Medium

76	Two Sum	Easy	leetcode.com/problems/two-sum
77	Sort an array of 0's 1's and 2's	Medium	leetcode.com/problems/sort-colors
78	Majority Element-I	Easy	leetcode.com/problems/majority-element
79	Kadane's Algorithm	Medium	leetcode.com/problems/maximum-subarray
80	Print subarray with maximum subarray sum (extended version of above problem)	Medium	no LeetCode · takeuforward.org
81	Stock Buy and Sell	Medium	leetcode.com/problems/best-time-to-buy-and-sell-stock
82	Rearrange array elements by sign	Medium	leetcode.com/problems/rearrange-array-elements-by-sign
83	Next Permutation	Medium	leetcode.com/problems/next-permutation
84	Leaders in an Array	Medium	no LeetCode · takeuforward.org
85	Longest Consecutive Sequence in an Array	Medium	leetcode.com/problems/longest-consecutive-sequence
86	Set Matrix Zeroes	Medium	leetcode.com/problems/set-matrix-zeroes
87	Rotate matrix by 90 degrees	Medium	leetcode.com/problems/rotate-image
88	Print the matrix in spiral manner	Medium	leetcode.com/problems/spiral-matrix

#	PROBLEM	LEVEL	LEETCODE LINK
89	Count subarrays with given sum	Medium	leetcode.com/problems/subarray-sum-equals-k
Hard			
90	Pascal's Triangle I	Easy	leetcode.com/problems/pascals-triangle
91	Majority Element-II	Hard	leetcode.com/problems/majority-element-ii
92	3 Sum	Medium	leetcode.com/problems/3sum
93	4 Sum	Medium	leetcode.com/problems/4sum
94	Largest Subarray with Sum 0	Medium	no LeetCode · takeuforward.org
95	Count subarrays with given xor K	Hard	no LeetCode · takeuforward.org
96	Merge Overlapping Subintervals	Medium	leetcode.com/problems/merge-intervals
97	Merge two sorted arrays without extra space	Medium	leetcode.com/problems/merge-sorted-array
98	Find the repeating and missing number	Hard	no LeetCode · takeuforward.org
99	Count Inversions	Hard	no LeetCode · takeuforward.org
100	Reverse Pairs	Hard	leetcode.com/problems/reverse-pairs
101	Maximum Product Subarray in an Array	Hard	leetcode.com/problems/maximum-product-subarray

Binary Search [1D, 2D Arrays, Search Space]

32 problems

#	PROBLEM	LEVEL	LEETCODE LINK
BS on 1D Arrays			
102	Search X in sorted array	Easy	leetcode.com/problems/binary-search
103	Lower Bound	Easy	no LeetCode · takeuforward.org
104	Upper Bound	Easy	no LeetCode · takeuforward.org
105	Search insert position	Easy	leetcode.com/problems/search-insert-position
106	Floor and Ceil in Sorted Array	Easy	no LeetCode · takeuforward.org
107	First and last occurrence	Easy	leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array
108	Count Occurrences in a Sorted Array	Easy	no LeetCode · takeuforward.org
109	Search in rotated sorted array-I	Medium	leetcode.com/problems/search-in-rotated-sorted-array
110	Search in rotated sorted array-II	Medium	leetcode.com/problems/search-in-rotated-sorted-array-ii
111	Find minimum in Rotated Sorted Array	Easy	leetcode.com/problems/find-minimum-in-rotated-sorted-array
112	Find out how many times the array is rotated	Easy	no LeetCode · takeuforward.org
113	Single element in a Sorted Array	Medium	leetcode.com/problems/single-element-in-a-sorted-array
114	Find peak element	Medium	leetcode.com/problems/find-peak-element
BS on Answers			
115	Find square root of a number	Medium	no LeetCode · takeuforward.org
116	Find Nth root of a number	Medium	no LeetCode · takeuforward.org
117	Koko eating bananas	Medium	leetcode.com/problems/koko-eating-bananas
118	Minimum days to make M bouquets	Medium	leetcode.com/problems/minimum-number-of-days-to-make-m-bouquets
119	Find the smallest divisor	Medium	leetcode.com/problems/find-the-smallest-divisor-given-a-threshold

#	PROBLEM	LEVEL	LEETCODE LINK
120	Capacity to Ship Packages Within D Days	Medium	leetcode.com/problems/capacity-to-ship-packages-within-d-days
121	Kth Missing Positive Number	Medium	leetcode.com/problems/kth-missing-positive-number
122	Aggressive Cows	Hard	no LeetCode · takeuforward.org
123	Book Allocation Problem	Hard	no LeetCode · takeuforward.org
124	Split array - largest sum	Hard	leetcode.com/problems/split-array-largest-sum
125	Painter's Partition	Medium	no LeetCode · takeuforward.org
126	Minimize Max Distance to Gas Station	Hard	leetcode.com/problems/minimize-max-distance-to-gas-station (Premium)
127	Median of 2 sorted arrays	Hard	leetcode.com/problems/median-of-two-sorted-arrays
128	Kth element of 2 sorted arrays	Medium	no LeetCode · takeuforward.org

BS on 2D Arrays

129	Find row with maximum 1's	Easy	no LeetCode · takeuforward.org
130	Search in a 2D matrix	Hard	leetcode.com/problems/search-a-2d-matrix
131	Search in 2D matrix - II	Hard	leetcode.com/problems/search-a-2d-matrix-ii
132	Find Peak Element - II	Medium	leetcode.com/problems/find-a-peak-element-ii
133	Matrix Median	Hard	no LeetCode · takeuforward.org

Strings [Basic and Medium]

15 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Basic and Easy String Problems			
134	Remove Outermost Parentheses	Medium	leetcode.com/problems/remove-outermost-parentheses
135	Reverse words in a given string / Palindrome Check	Medium	leetcode.com/problems/reverse-words-in-a-string
136	Largest Odd Number in a String	Easy	leetcode.com/problems/largest-odd-number-in-string
137	Longest Common Prefix	Easy	leetcode.com/problems/longest-common-prefix
138	Isomorphic String	Easy	leetcode.com/problems/isomorphic-strings
139	Rotate String	Easy	leetcode.com/problems/rotate-string
140	Check if two strings are anagram of each other	Easy	leetcode.com/problems/valid-anagram

Medium String Problems

141	Sort Characters by Frequency	Easy	leetcode.com/problems/sort-characters-by-frequency
142	Maximum Nesting Depth of the Parentheses	Medium	leetcode.com/problems/maximum-nesting-depth-of-the-parentheses
143	Roman to Integer	Medium	leetcode.com/problems/roman-to-integer
144	String to Integer (atoi)	Medium	leetcode.com/problems/string-to-integer-atoi
145	Count Number of Substrings	Easy	no LeetCode · takeuforward.org
146	Longest Palindromic Substring	Medium	leetcode.com/problems/longest-palindromic-substring
147	Sum of Beauty of All Substrings	Medium	leetcode.com/problems/sum-of-beauty-of-all-substrings
148	Reverse every word in a string	Medium	leetcode.com/problems/reverse-words-in-a-string

Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]

31 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Learn 1D LinkedList			
149	Introduction to Singly LinkedList	Easy	no LeetCode · takeuforward.org
150	Insertion at the head of Linked List	Easy	no LeetCode · takeuforward.org
151	Deletion of the head of LL	Easy	leetcode.com/problems/delete-node-in-a-linked-list
152	Find the length of the Linked List	Easy	no LeetCode · takeuforward.org
153	Search in Linked List	Medium	no LeetCode · takeuforward.org
Learn Doubly LinkedList			
154	Introduction to Doubly LL	Easy	no LeetCode · takeuforward.org
155	Insert node before head in Doubly Linked List	Easy	no LeetCode · takeuforward.org
156	Delete head of Doubly Linked List	Easy	no LeetCode · takeuforward.org
157	Reverse a Doubly Linked List	Medium	no LeetCode · takeuforward.org
Medium Problems of LL			
158	Middle of a LinkedList [TortoiseHare Method]	Easy	leetcode.com/problems/middle-of-the-linked-list
159	Reverse a LinkedList [Iterative]	Medium	leetcode.com/problems/reverse-linked-list
160	Reverse a LL	Medium	leetcode.com/problems/reverse-linked-list
161	Detect a loop in LL	Medium	leetcode.com/problems/linked-list-cycle
162	Find the starting point in LL	Medium	leetcode.com/problems/linked-list-cycle-ii
163	Length of loop in LL	Medium	no LeetCode · takeuforward.org
164	Check if LL is palindrome or not	Medium	leetcode.com/problems/palindrome-linked-list
165	Segregate odd and even nodes in Linked List	Medium	leetcode.com/problems/odd-even-linked-list
166	Remove Nth node from the back of the LL	Medium	leetcode.com/problems/remove-nth-node-from-end-of-list
167	Delete the middle node in LL	Medium	leetcode.com/problems/delete-the-middle-node-of-a-linked-list
168	Sort LL	Hard	leetcode.com/problems/sort-list
169	Sort a Linked List of 0's 1's and 2's	Medium	no LeetCode · takeuforward.org
170	Find the intersection point of Y LL	Medium	leetcode.com/problems/intersection-of-two-linked-lists
171	Add one to a number represented by LL	Medium	no LeetCode · takeuforward.org
172	Add two numbers in Linked List	Medium	leetcode.com/problems/add-two-numbers
Medium Problems of DLL			
173	Delete all occurrences of a key in DLL	Hard	no LeetCode · takeuforward.org
174	Find Pairs with Given Sum in Doubly Linked List	Medium	no LeetCode · takeuforward.org
175	Remove duplicates from sorted DLL	Hard	no LeetCode · takeuforward.org
Hard Problems of LL			
176	Reverse LL in group of given size K	Hard	leetcode.com/problems/reverse-nodes-in-k-group
177	Rotate a LL	Hard	leetcode.com/problems/rotate-list
178	Flattening of LL	Hard	no LeetCode · takeuforward.org
179	Clone a LL with random and next pointer	Hard	leetcode.com/problems/copy-list-with-random-pointer

Recursion [PatternWise]

25 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Get a Strong Hold			
180	Recursive Implementation of atoi()	Medium	leetcode.com/problems/string-to-integer-atoi
181	Pow(x, n)	Easy	leetcode.com/problems/powx-n
182	Count Good Numbers	Medium	leetcode.com/problems/count-good-numbers
183	Sort a stack using recursion	Medium	no LeetCode · takeuforward.org
184	Reverse a Stack	Medium	no LeetCode · takeuforward.org
Subsequences Pattern			
185	Generate Binary Strings Without Consecutive 1s	Medium	no LeetCode · takeuforward.org
186	Generate Parentheses	Medium	leetcode.com/problems/generate-parentheses
187	Power Set	Medium	no LeetCode · takeuforward.org
188	Learn All Patterns of Subsequences (Theory)	Easy	no LeetCode · takeuforward.org
189	Count all subsequences with sum K	Easy	no LeetCode · takeuforward.org
190	Check if there exists a subsequence with sum K	Easy	no LeetCode · takeuforward.org
191	Combination Sum	Medium	leetcode.com/problems/combination-sum
192	Combination Sum II	Medium	leetcode.com/problems/combination-sum-ii
193	Subsets I	Medium	no LeetCode · takeuforward.org
194	Subsets II	Medium	leetcode.com/problems/subsets-ii
195	Combination Sum III	Medium	leetcode.com/problems/combination-sum-iii
196	Letter Combinations of a Phone Number	Hard	leetcode.com/problems/letter-combinations-of-a-phone-number
Trying out all Combos / Hard			
197	Palindrome partitioning	Hard	no LeetCode
198	Word Search	Hard	leetcode.com/problems/word-search
199	N Queen	Hard	leetcode.com/problems/n-queens
200	Rat in a Maze	Hard	no LeetCode · takeuforward.org
201	Word Break	Medium	no LeetCode
202	M Coloring Problem	Hard	no LeetCode · takeuforward.org
203	Sudoku Solver	Hard	leetcode.com/problems/sudoku-solver
204	Expression Add Operators	Hard	leetcode.com/problems/expression-add-operators

Bit Manipulation [Concepts & Problems]

18 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Learn Bit Manipulation			
205	Introduction to Bits and Tricks	Easy	no LeetCode · takeuforward.org
206	Check if the i-th bit is Set or Not	Easy	no LeetCode · takeuforward.org
207	Check if a Number is Odd or Not	Easy	no LeetCode · takeuforward.org
208	Check if a Number is Power of 2 or Not	Easy	leetcode.com/problems/power-of-two
209	Count the Number of Set Bits	Easy	no LeetCode · takeuforward.org

#	PROBLEM	LEVEL	LEETCODE LINK
210	Set/Unset the rightmost unset bit	Easy	no LeetCode · takeuforward.org
211	Swap Two Numbers	Easy	no LeetCode · takeuforward.org
212	Divide two numbers without multiplication and division	Medium	leetcode.com/problems/divide-two-integers

Interview Problems

213	Minimum Bit Flips to Convert Number	Medium	leetcode.com/problems/minimum-bit-flips-to-convert-number
214	Single Number - I	Medium	leetcode.com/problems/single-number
215	Power Set Bit Manipulation	Medium	leetcode.com/problems/subsets
216	XOR of numbers in a given range	Medium	no LeetCode · takeuforward.org
217	Single Number - III	Medium	no LeetCode · takeuforward.org

Advanced Maths

218	Print Prime Factors of a Number	Hard	no LeetCode
219	Divisors of a Number	Easy	no LeetCode · takeuforward.org
220	Count primes in range L to R	Hard	leetcode.com/problems/count-primes
221	Prime factorisation of a Number	Hard	no LeetCode
222	Pow(x,n)	Medium	leetcode.com/problems/powx-n

Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]

30 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Learning			
223	Implement Stack using Arrays	Easy	no LeetCode · takeuforward.org
224	Implement Queue using Arrays	Easy	no LeetCode · takeuforward.org
225	Implement Stack using Queue	Easy	leetcode.com/problems/implement-stack-using-queues
226	Implement Queue using Stack	Easy	leetcode.com/problems/implement-queue-using-stacks
227	Implement stack using Linkedlist	Easy	no LeetCode · takeuforward.org
228	Implement queue using Linkedlist	Easy	no LeetCode · takeuforward.org
229	Balanced Paranthesis	Easy	leetcode.com/problems/valid-parentheses
230	Implement Min Stack	Hard	leetcode.com/problems/min-stack

Prefix, Infix, PostFix Conversion Problems

231	Infix to Postfix Conversion	Medium	no LeetCode · takeuforward.org
232	Prefix to Infix Conversion	Medium	no LeetCode · takeuforward.org
233	Prefix to Postfix Conversion	Medium	no LeetCode · takeuforward.org
234	Postfix to Prefix Conversion	Medium	no LeetCode · takeuforward.org
235	Postfix to Infix Conversion	Easy	no LeetCode · takeuforward.org
236	Infix to Prefix Conversion	Medium	no LeetCode · takeuforward.org

Monotonic Stack/Queue Problems [VVV. Imp]

237	Next Greater Element	Medium	leetcode.com/problems/next-greater-element-i
238	Next Greater Element - 2	Medium	leetcode.com/problems/next-greater-element-ii
239	Next Smaller Element	Medium	no LeetCode · takeuforward.org

#	PROBLEM	LEVEL	LEETCODE LINK
240	Number of Greater Elements to the Right	Easy	no LeetCode · takeuforward.org
241	Trapping Rainwater	Hard	leetcode.com/problems/trapping-rain-water
242	Sum of Subarray Minimums	Medium	leetcode.com/problems/sum-of-subarray-minimums
243	Asteroid Collision	Medium	leetcode.com/problems/asteroid-collision
244	Sum of Subarray Ranges	Medium	leetcode.com/problems/sum-of-subarray-ranges
245	Remove K Digits	Medium	leetcode.com/problems/remove-k-digits
246	Largest rectangle in a histogram	Hard	leetcode.com/problems/largest-rectangle-in-histogram
247	Maximum Rectangles	Hard	leetcode.com/problems/maximal-rectangle

Implementation Problems

248	Sliding Window Maximum	Hard	leetcode.com/problems/sliding-window-maximum
249	Stock span problem	Hard	leetcode.com/problems/online-stock-span
250	Celebrity Problem	Hard	leetcode.com/problems/find-the-celebrity (Premium)
251	LRU Cache	Medium	no LeetCode · takeuforward.org
252	LFU Cache	Hard	leetcode.com/problems/lfu-cache

Sliding Window & Two Pointer Combined Problems

12 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Medium Problems			
253	Longest Substring Without Repeating Characters	Medium	leetcode.com/problems/longest-substring-without-repeating-characters
254	Max Consecutive Ones III	Medium	leetcode.com/problems/max-consecutive-ones-iii
255	Fruit Into Baskets	Medium	leetcode.com/problems/fruit-into-baskets
256	Longest Repeating Character Replacement	Hard	leetcode.com/problems/longest-repeating-character-replacement
257	Binary Subarrays With Sum	Hard	leetcode.com/problems/binary-subarrays-with-sum
258	Count number of Nice subarrays	Hard	leetcode.com/problems/count-number-of-nice-subarrays
259	Number of Substrings Containing All Three Characters	Hard	leetcode.com/problems/number-of-substrings-containing-all-three-characters
260	Maximum Points You Can Obtain from Cards	Medium	leetcode.com/problems/maximum-points-you-can-obtain-from-cards

Hard Problems

261	Longest Substring With At Most K Distinct Characters	Hard	leetcode.com/problems/longest-substring-with-at-most-k-distinct-characters (Premium)
262	Subarrays with K Different Integers	Medium	leetcode.com/problems/subarrays-with-k-different-integers
263	Minimum Window Substring	Hard	leetcode.com/problems/minimum-window-substring
264	Minimum Window Subsequence	Hard	leetcode.com/problems/minimum-window-subsequence (Premium)

Heaps [Learning, Medium, Hard Problems]

17 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Learning			

#	PROBLEM	LEVEL	LEETCODE LINK
265	Heaps (Theory Video)	Easy	no LeetCode · takeuforward.org
266	Implement Min Heap	Medium	no LeetCode
267	Check if an array represents a min heap	Medium	no LeetCode · takeuforward.org
268	Convert Min Heap to Max Heap	Medium	no LeetCode

Medium Problems

269	K-th Largest element in an array	Medium	no LeetCode · takeuforward.org
270	Kth smallest element in an array [use priority queue]	Medium	no LeetCode · takeuforward.org
271	Sort K sorted array	Easy	no LeetCode · takeuforward.org
272	Merge K sorted Lists	Hard	leetcode.com/problems/merge-k-sorted-lists
273	Replace Elements by Their Rank	Easy	no LeetCode · takeuforward.org
274	Task Scheduler	Medium	leetcode.com/problems/task-scheduler
275	Hand of Straights	Medium	leetcode.com/problems/hand-of-straights

Hard Problems

276	Design Twitter	Medium	leetcode.com/problems/design-twitter
277	Minimum Cost to Connect Sticks	Medium	no LeetCode · takeuforward.org
278	Kth largest element in a stream of running integers	Hard	leetcode.com/problems/kth-largest-element-in-a-stream
279	Maximum Sum Combination	Hard	no LeetCode · takeuforward.org
280	Find Median from Data Stream	Hard	leetcode.com/problems/find-median-from-data-stream
281	Top K Frequent Elements	Medium	leetcode.com/problems/top-k-frequent-elements

Greedy Algorithms [Easy, Medium/Hard]

15 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Easy Problems			
282	Assign Cookies	Easy	leetcode.com/problems/assign-cookies
283	Fractional Knapsack	Medium	no LeetCode · takeuforward.org
284	Lemonade Change	Easy	leetcode.com/problems/lemonade-change
285	Valid Paranthesis Checker	Hard	leetcode.com/problems/valid-parenthesis-string
Medium/Hard			
286	N meetings in one room	Medium	no LeetCode · takeuforward.org
287	Jump Game - I	Easy	leetcode.com/problems/jump-game
288	Jump Game II	Medium	leetcode.com/problems/jump-game-ii
289	Minimum number of platforms required for a railway	Medium	no LeetCode · takeuforward.org
290	Job sequencing Problem	Medium	no LeetCode · takeuforward.org
291	Candy	Hard	leetcode.com/problems/candy
292	Shortest Job First	Medium	no LeetCode · takeuforward.org
293	Program for Least Recently Used (LRU) Page Replacement Algorithm	Medium	no LeetCode · takeuforward.org
294	Insert Interval	Medium	leetcode.com/problems/insert-interval
295	Merge Intervals	Medium	leetcode.com/problems/merge-intervals

#	PROBLEM	LEVEL	LEETCODE LINK
296	Non-overlapping Intervals	Medium	leetcode.com/problems/non-overlapping-intervals

Binary Trees [Traversals, Medium and Hard Problems]

38 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Traversals			
297	Introduction to Trees	Easy	no LeetCode · takeuforward.org
298	Binary Tree Representation in Java	Easy	no LeetCode · takeuforward.org
299	Pre, Post, Inorder in one traversal	Easy	no LeetCode · takeuforward.org
300	Preorder Traversal	Easy	leetcode.com/problems/binary-tree-preorder-traversal
301	Inorder Traversal of Binary Tree	Easy	leetcode.com/problems/binary-tree-inorder-traversal
302	Postorder Traversal	Easy	leetcode.com/problems/binary-tree-postorder-traversal
303	Level Order Traversal	Easy	leetcode.com/problems/binary-tree-level-order-traversal
304	Iterative Preorder Traversal of Binary Tree	Easy	leetcode.com/problems/binary-tree-preorder-traversal
305	Iterative Inorder Traversal of Binary Tree	Easy	leetcode.com/problems/binary-tree-inorder-traversal
306	Post-order Traversal of Binary Tree using 2 stack	Easy	leetcode.com/problems/binary-tree-postorder-traversal
307	Post-order Traversal of Binary Tree using 1 stack	Easy	leetcode.com/problems/binary-tree-postorder-traversal
308	Preorder, Inorder, and Postorder Traversal in one Traversal	Easy	no LeetCode · takeuforward.org
Medium Problems			
309	Maximum Depth in BT	Medium	leetcode.com/problems/maximum-depth-of-binary-tree
310	Check for balanced binary tree	Medium	leetcode.com/problems/balanced-binary-tree
311	Diameter of Binary Tree	Easy	leetcode.com/problems/diameter-of-binary-tree
312	Maximum path sum	Medium	leetcode.com/problems/binary-tree-maximum-path-sum
313	Check if two trees are identical or not	Medium	leetcode.com/problems/same-tree
314	Zig Zag or Spiral Traversal	Medium	leetcode.com/problems/binary-tree-zigzag-level-order-traversal
315	Boundary Traversal	Medium	leetcode.com/problems/boundary-of-binary-tree (Premium)
316	Vertical Order Traversal	Medium	leetcode.com/problems/vertical-order-traversal-of-a-binary-tree
317	Top View of BT	Medium	no LeetCode · takeuforward.org
318	Bottom view of BT	Medium	no LeetCode · takeuforward.org
319	Right/Left View of Binary Tree	Medium	leetcode.com/problems/binary-tree-right-side-view
320	Symmetric Binary Tree	Medium	leetcode.com/problems/symmetric-tree
Hard Problems			
321	Print root to leaf path in BT	Medium	no LeetCode · takeuforward.org
322	LCA in BT	Hard	leetcode.com/problems/lowest-common-ancestor-of-a-binary-tree
323	Maximum Width of BT	Medium	leetcode.com/problems/maximum-width-of-binary-tree
324	Children Sum Property in Binary Tree	Medium	no LeetCode · takeuforward.org
325	Print all nodes at a distance of K in BT	Hard	leetcode.com/problems/all-nodes-distance-k-in-binary-tree
326	Minimum time taken to burn the BT from a given Node	Hard	no LeetCode · takeuforward.org
327	Count total nodes in a complete BT	Easy	leetcode.com/problems/count-complete-tree-nodes

#	PROBLEM	LEVEL	LEETCODE LINK
328	Requirements needed to construct a unique BT	Medium	no LeetCode
329	Construct a BT from Preorder and Inorder	Hard	leetcode.com/problems/construct-binary-tree-from-preorder-and-inorder-traversal
330	Construct the Binary Tree from Postorder and Inorder Traversal	Hard	leetcode.com/problems/construct-binary-tree-from-inorder-and-postorder-traversal
331	Serialize and De-serialize BT	Hard	leetcode.com/problems/serialize-and-deserialize-binary-tree
332	Morris Preorder Traversal of a Binary Tree	Hard	leetcode.com/problems/binary-tree-inorder-traversal
333	Morris Inorder Traversal of a Binary Tree	Hard	leetcode.com/problems/binary-tree-inorder-traversal
334	Flatten Binary Tree to Linked List	Medium	leetcode.com/problems/flatten-binary-tree-to-linked-list

Binary Search Trees [Concept and Problems]

16 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Concepts			
335	Introduction to BST	Easy	no LeetCode · takeuforward.org
336	Search in a Binary Search Tree	Easy	leetcode.com/problems/search-in-a-binary-search-tree
337	Find Min/Max in BST	Easy	no LeetCode · takeuforward.org
Practice Problems			
338	Floor and Ceil in a BST	Easy	no LeetCode
339	Floor in a Binary Search Tree	Easy	no LeetCode · takeuforward.org
340	Insert a given node in BST	Medium	leetcode.com/problems/insert-into-a-binary-search-tree
341	Delete a node in BST	Medium	leetcode.com/problems/delete-node-in-a-bst
342	Kth Smallest and Largest element in BST	Medium	leetcode.com/problems/kth-smallest-element-in-a-bst
343	Check if a tree is a BST or not	Medium	leetcode.com/problems/validate-binary-search-tree
344	LCA in BST	Medium	leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree
345	Construct a BST from a preorder traversal	Medium	leetcode.com/problems/construct-binary-search-tree-from-preorder-traversal
346	Inorder Successor/Predecessor in BST	Medium	leetcode.com/problems/inorder-successor-in-bst (Premium)
347	Merge 2 BST's	Hard	leetcode.com/problems/binary-search-tree-iterator
348	Two Sum In BST Check if there exists a pair with Sum K	Hard	leetcode.com/problems/two-sum-iv-input-is-a-bst
349	Correct BST with two nodes swapped	Hard	leetcode.com/problems/recover-binary-search-tree
350	Largest BST in Binary Tree	Hard	leetcode.com/problems/maximum-sum-bst-in-binary-tree

Graphs [Concepts & Problems]

53 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Learning			
351	Introduction to Graph	Easy	no LeetCode · takeuforward.org
352	Graph Representation C++	Easy	no LeetCode · takeuforward.org
353	Graph Representation Java	Easy	no LeetCode · takeuforward.org
354	Connected Components	Medium	no LeetCode · takeuforward.org

#	PROBLEM	LEVEL	LEETCODE LINK
355	Traversal Techniques	Medium	no LeetCode · takeuforward.org
356	DFS	Medium	no LeetCode · takeuforward.org
Problems on BFS/DFS			
357	Number of provinces	Medium	leetcode.com/problems/number-of-provinces
358	Connected Components Problem in Matrix	Medium	no LeetCode · takeuforward.org
359	Rotten Oranges	Medium	leetcode.com/problems/rotting-oranges
360	Flood fill algorithm	Medium	leetcode.com/problems/flood-fill
361	Cycle Detection in Undirected Graph (bfs)	Hard	no LeetCode · takeuforward.org
362	Detect a cycle in an undirected graph	Hard	leetcode.com/problems/course-schedule
363	Distance of nearest cell having one	Medium	leetcode.com/problems/01-matrix
364	Surrounded Regions	Medium	leetcode.com/problems/surrounded-regions
365	Number of enclaves	Medium	leetcode.com/problems/number-of-enclaves
366	Word ladder I	Hard	leetcode.com/problems/word-ladder
367	Word ladder II	Hard	leetcode.com/problems/word-ladder-ii
368	Number of islands	Medium	leetcode.com/problems/number-of-islands
369	Bipartite Graph (DFS)	Hard	leetcode.com/problems/is-graph-bipartite
370	Cycle Detection in Directed Graph (DFS)	Hard	leetcode.com/problems/course-schedule-ii
Topo Sort and Problems			
371	Topo Sort	Hard	no LeetCode · takeuforward.org
372	Topological sort or Kahn's algorithm	Hard	no LeetCode · takeuforward.org
373	Detect a cycle in a directed graph	Hard	leetcode.com/problems/course-schedule
374	Course Schedule I	Hard	leetcode.com/problems/course-schedule
375	Course Schedule II	Medium	leetcode.com/problems/course-schedule-ii
376	Find eventual safe states	Hard	leetcode.com/problems/find-eventual-safe-states
377	Alien Dictionary	Hard	leetcode.com/problems/alien-dictionary (Premium)
Shortest Path Algorithms and Problems			
378	Shortest path in undirected graph with unit weights	Hard	no LeetCode · takeuforward.org
379	Shortest path in DAG	Hard	no LeetCode · takeuforward.org
380	Dijkstra's Algorithm	Hard	no LeetCode · takeuforward.org
381	Why priority Queue is used in Dijkstra's Algorithm	Hard	no LeetCode · takeuforward.org
382	Shortest Distance in a Binary Maze	Hard	leetcode.com/problems/shortest-path-in-binary-matrix
383	Path with minimum effort	Hard	leetcode.com/problems/path-with-minimum-effort
384	Cheapest flight within K stops	Hard	leetcode.com/problems/cheapest-flights-within-k-stops
385	Network Delay Time	Medium	leetcode.com/problems/network-delay-time
386	Number of ways to arrive at destination	Hard	leetcode.com/problems/number-of-ways-to-arrive-at-destination
387	Minimum multiplications to reach end	Hard	no LeetCode · takeuforward.org
388	Bellman Ford Algorithm	Hard	no LeetCode · takeuforward.org
389	Floyd warshall algorithm	Hard	no LeetCode · takeuforward.org
390	Find the city with the smallest number of neighbors	Hard	leetcode.com/problems/find-the-city-with-the-smallest-number-of-neighbors-at-a-threshold-distance

#	PROBLEM	LEVEL	LEETCODE LINK
MinimumSpanningTree/Disjoint Set and Problems			
391	MST theory	Easy	no LeetCode · takeuforward.org
392	Prim's Algorithm	Hard	no LeetCode · takeuforward.org
393	Disjoint Set	Hard	no LeetCode · takeuforward.org
394	Find the MST weight	Hard	no LeetCode · takeuforward.org
395	Number of operations to make network connected	Hard	leetcode.com/problems/number-of-operations-to-make-network-connected
396	Most stones removed with same row or column	Medium	leetcode.com/problems/most-stones-removed-with-same-row-or-column
397	Accounts merge	Hard	leetcode.com/problems/accounts-merge
398	Number of islands II	Hard	leetcode.com/problems/number-of-islands-ii (Premium)
399	Making a large island	Hard	leetcode.com/problems/making-a-large-island
400	Swim in Rising Water	Medium	leetcode.com/problems/swim-in-rising-water

Other Algorithms

401	Bridges in graph	Hard	leetcode.com/problems/critical-connections-in-a-network
402	Articulation point in graph	Hard	no LeetCode · takeuforward.org
403	Kosaraju's algorithm	Hard	no LeetCode

Dynamic Programming [Patterns and Problems]

55 problems

#	PROBLEM	LEVEL	LEETCODE LINK
Introduction to DP			
404	Introduction to DP	Easy	no LeetCode · takeuforward.org
1D DP			
405	Climbing stairs	Medium	leetcode.com/problems/climbing-stairs
406	Frog Jump	Medium	no LeetCode · takeuforward.org
407	Frog jump with K distances	Medium	no LeetCode · takeuforward.org
408	Maximum sum of non adjacent elements	Medium	leetcode.com/problems/house-robber
409	House robber	Medium	leetcode.com/problems/house-robber-ii
2D/3D DP and DP on Grids			
410	Ninja's training	Medium	no LeetCode · takeuforward.org
411	Grid Unique Paths : DP on Grids (DP8)	Medium	leetcode.com/problems/unique-paths
412	Unique paths II	Medium	leetcode.com/problems/unique-paths-ii
413	Minimum Falling Path Sum	Medium	leetcode.com/problems/minimum-path-sum
414	Triangle	Medium	leetcode.com/problems/triangle
415	Ninja and his Friends	Medium	no LeetCode · takeuforward.org
DP on Subsequences			
416	Subset sum equal to target (DP- 14)	Hard	no LeetCode · takeuforward.org
417	Partition equal subset sum	Hard	leetcode.com/problems/partition-equal-subset-sum
418	Partition a set into two subsets with minimum absolute sum difference	Hard	leetcode.com/problems/partition-array-into-two-arrays-to-minimize-sum-difference

#	PROBLEM	LEVEL	LEETCODE LINK
419	Count subsets with sum K	Hard	no LeetCode · takeuforward.org
420	Count partitions with given difference	Hard	no LeetCode · takeuforward.org
421	Assign Cookies	Easy	leetcode.com/problems/assign-cookies
422	Minimum Coins (DP - 20)	Hard	leetcode.com/problems/coin-change
423	Target sum	Hard	leetcode.com/problems/target-sum
424	Coin Change 2 (DP - 22)	Hard	leetcode.com/problems/coin-change-2 (Premium)
425	Unbounded knapsack	Hard	no LeetCode · takeuforward.org
426	Rod Cutting Problem (DP - 24)	Hard	no LeetCode · takeuforward.org

DP on Strings

427	Longest common subsequence	Hard	no LeetCode · takeuforward.org
428	Print Longest Common Subsequence (DP - 26)	Hard	no LeetCode · takeuforward.org
429	Longest common substring	Hard	no LeetCode · takeuforward.org
430	Longest palindromic subsequence	Hard	leetcode.com/problems/longest-palindromic-subsequence
431	Minimum insertions to make string palindrome DP-29	Hard	leetcode.com/problems/minimum-insertion-steps-to-make-a-string-palindrome
432	Minimum insertions or deletions to convert string A to B	Hard	leetcode.com/problems/delete-operation-for-two-strings
433	Shortest common supersequence	Hard	leetcode.com/problems/shortest-common-supersequence
434	Distinct subsequences	Hard	leetcode.com/problems/distinct-subsequences
435	Edit distance	Hard	leetcode.com/problems/edit-distance
436	Wildcard matching	Hard	leetcode.com/problems/wildcard-matching

DP on Stocks

437	Best time to buy and sell stock	Medium	leetcode.com/problems/best-time-to-buy-and-sell-stock
438	Best time to buy and sell stock II	Medium	leetcode.com/problems/best-time-to-buy-and-sell-stock-ii
439	Best time to buy and sell stock III	Medium	leetcode.com/problems/best-time-to-buy-and-sell-stock-iii
440	Best time to buy and sell stock IV	Medium	leetcode.com/problems/best-time-to-buy-and-sell-stock-iv
441	Best Time to Buy and Sell Stock with Cooldown	Medium	leetcode.com/problems/best-time-to-buy-and-sell-stock-with-cooldown
442	Best time to buy and sell stock with transaction fees	Medium	leetcode.com/problems/best-time-to-buy-and-sell-stock-with-transaction-fee

DP on LIS

443	Longest Increasing Subsequence	Medium	no LeetCode · takeuforward.org
444	Print Longest Increasing Subsequence	Medium	no LeetCode · takeuforward.org
445	Longest Increasing Subsequence (DP-43)	Medium	no LeetCode · takeuforward.org
446	Largest Divisible Subset	Medium	leetcode.com/problems/largest-divisible-subset
447	Longest String Chain	Medium	leetcode.com/problems/longest-string-chain
448	Longest Bitonic Subsequence	Medium	no LeetCode · takeuforward.org
449	Number of Longest Increasing Subsequences	Medium	leetcode.com/problems/number-of-longest-increasing-subsequence

MCM DP | Partition DP

450	Matrix chain multiplication	Hard	no LeetCode · takeuforward.org
451	Matrix Chain Multiplication Bottom-Up (DP-49)	Hard	no LeetCode · takeuforward.org

#	PROBLEM	LEVEL	LEETCODE LINK
452	Minimum cost to cut the stick	Hard	leetcode.com/problems/minimum-cost-to-cut-a-stick
453	Burst balloons	Hard	leetcode.com/problems/burst-balloons
454	Different Ways to Evaluate a Boolean Expression	Medium	leetcode.com/problems/parsing-a-boolean-expression
455	Palindrome partitioning II	Hard	leetcode.com/problems/palindrome-partitioning-ii
456	Partition Array for Maximum Sum	Medium	leetcode.com/problems/partition-array-for-maximum-sum

DP on Squares

457	Maximum Rectangle Area with all 1's (DP-55)	Hard	leetcode.com/problems/maximal-rectangle
458	Count Square Submatrices with All Ones (DP-56)	Easy	leetcode.com/problems/count-square-submatrices-with-all-ones

Tries

7 problems

#	PROBLEM	LEVEL	LEETCODE LINK
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Theory

459	Trie Implementation and Operations	Hard	leetcode.com/problems/implement-trie-prefix-tree
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Problems

460	Trie Implementation and Advanced Operations	Hard	no LeetCode · takeuforward.org
461	Longest Word with All Prefixes	Medium	no LeetCode
462	Number of distinct substrings in a string	Medium	no LeetCode · takeuforward.org
463	Bit PreRequisites for TRIE Problems	Easy	no LeetCode
464	Maximum XOR of two numbers in an array	Hard	leetcode.com/problems/maximum-xor-of-two-numbers-in-an-array
465	Maximum Xor with an element from an array	Hard	leetcode.com/problems/maximum-xor-with-an-element-from-array

Strings

9 problems

#	PROBLEM	LEVEL	LEETCODE LINK
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Hard Problems

466	Minimum number of bracket reversals to make an expression balanced	Hard	leetcode.com/problems/minimum-add-to-make-parentheses-valid
467	Count and say	Hard	leetcode.com/problems/count-and-say
468	Hashing In Strings Theory	Easy	no LeetCode · takeuforward.org
469	Rabin Karp Algorithm	Hard	no LeetCode
470	Z function	Hard	no LeetCode
471	KMP Algorithm or LPS array	Hard	leetcode.com/problems/find-the-index-of-the-first-occurrence-in-a-string
472	Shortest Palindrome	Hard	no LeetCode
473	Longest happy prefix	Hard	leetcode.com/problems/longest-happy-prefix
474	Count Palindromic Subsequences	Medium	leetcode.com/problems/count-palindromic-subsequences